Kingdom of Saudi Arabia
Ministry of Higher Education
Al-Imam Muhammad Ibn Saud Islamic University
College of Computer and Information Sciences

Examining Information Systems for Improved Patient Management in Healthcare (King Fahd Medical City Hospital)



IS1100

Presentated by

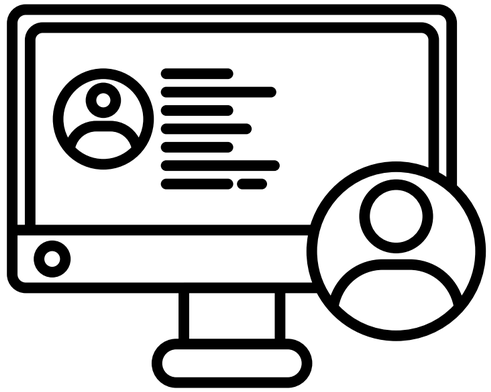
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1. Project introduction:



- About visiting King Fahad Medical City Hospital.
- How was the meeting with Engineer Hessa Al-Juhaiman coordinated?
- Digital transformation in the hospital: Moving from paper systems to digital systems.

1. Project introduction:



- How can senior management benefit from organizing the workflow?
- Display the current system in use and future upgrade plans for the system.
- The role of information systems in solving challenges.
- The need to adopt modern systems to deal with large amounts of data.

Role of the system :

- Security and data encryption.
- Manages patients’helth records
- Enhance communicatio:

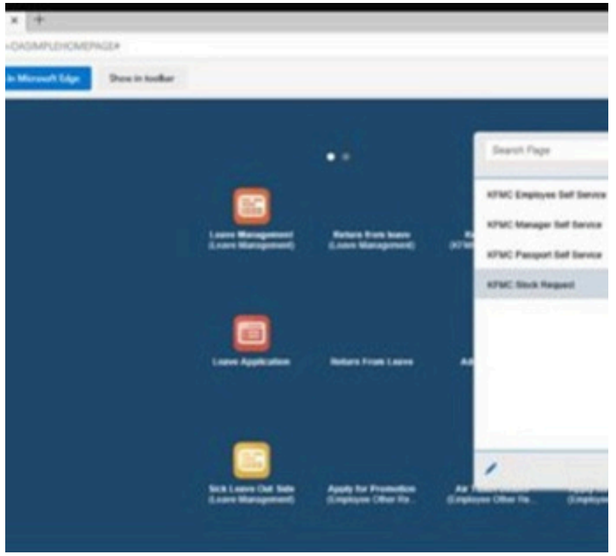
1/Facilitating communication between doctors and patients.

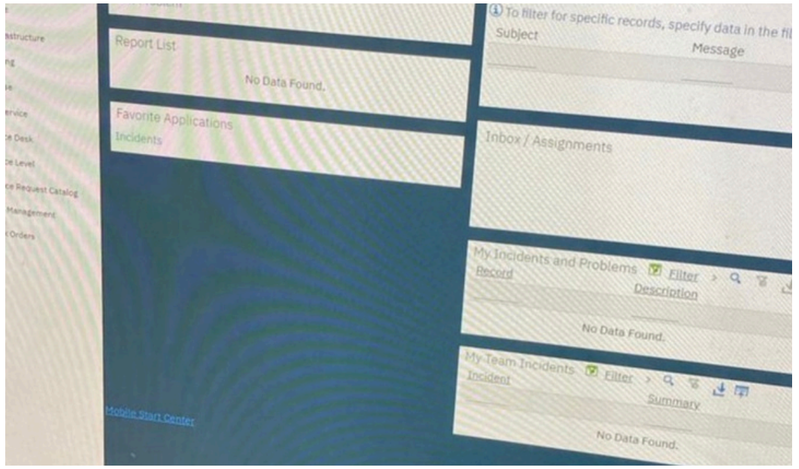
2/ Facilitating communication between doctors and pharmacists.

- Direct access to information.
- Direct update.
- Alerts and Notifications.

System Features:

- Easy-to-use interface .
- User traning.
- Supports mobile devices.
- Update information immediately upon entering it.
- Save time.
- Access via the Internet.
- Quick browsing.
- Data protection.
- Reducing medical errors.
- Financial operations management.





Description of stakeholders and business needs:



•The hospital was selected based on the importance of stakeholders.

• Pharmacies are also stakeholders: to inventory medicines, know the dates, and identify the beneficiaries.

• Doctors: Organize schedules and appointments, and easily connect patients with doctors.

Technical Details

The multiple systems used by King Fahad Medical City (KFMC) include

Electronic Patient Information Charting (EPIC)
Resource Management System (RMS)
Manage Engine, which is used for engineers and request submitters,
.is in the process of being replaced by another system called Maximum

Manage Engine is used to provide technical support in case of any
.malfunction

Driving Factors

The use of advanced technologies helps foster more innovation and technological development, such as KFMC’s use of EPIC and RMS. These systems give the hospital a wide reputation, making it a trusted healthcare provider.

Restraining Factors

The hospital faces some financial constraints and economic challenges, especially when it comes to providing communication devices like switches, which are important and costly.

.Changes in government regulations may also cause some issues for the hospital

Despite the hospital’s use of many advanced systems, there are still some older .programs that come with many limitations

Current Issue with the Information System

There is difficulty in linking the hospital with its external health centers due to the lack of sufficient switches.

The problem can be solved by assigning a person in each department responsible for quickly transferring and receiving information, troubleshooting, and providing more switches. This person should also be able to request new devices if there is a shortage or if .existing equipment is not efficient

limitations.

1- Data quality

2- Training and development cost

3- Resistance to change

4- Multiple systems

System methodology.

Data analysis :
should identify gaps and problems associated with these systems.

Defining requirements:
We must accurately determine the requirements and needs of employees and customers.

System design.
Design the system based on specific needs, with remote access and improved integration between the systems used.



System methodology.

System implementation.

Then implement the system and provide intensive training to all employees.

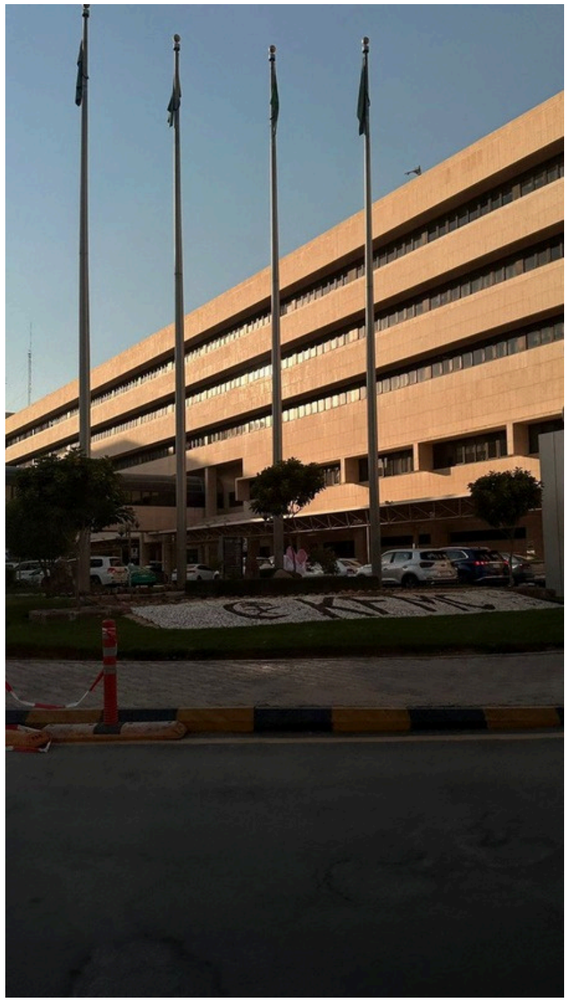
System Test.

System testing before deployment.



Project Experience Description

• Selected King Fahad Medical City as the project’s focus.
• Encountered delays in response and scheduling challenges.
• Conducted data analysis and proposed effective solutions.
• The experience enhanced our analytical and problem-solving capabilities.



Conclusion

At the end of our project on the hospital visit and review of the information system used in it, we can conclude that this experience was a valuable learning experience. We had the opportunity to learn how medical information is managed and how the system contributes to improving the quality of healthcare.